Marcos Ibáñez Fandos

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ABOUT ME

Engineering student with a strong interest in applied technologies, data analysis, and AI-based decision systems. Enjoys working between technical and business contexts, contributing to the development of products and collaborating with different profiles to improve how data is used in practice.

WORK EXPERIENCE

BMW Group (Internship) – Munich

Transaction Pricing

[16 Mar 2026 – Current]

Pamesa Ceramics (Internship) – Vila-real

Website: www.pamesa.com | **Business or sector:** Manufacturing

Network Infrastructure Technician

[1 Jun 2023 – 31 Aug 2023]

- Deployment and maintenance of network infrastructure and structured cabling.
- Installation, configuration, and management of racks in corporate environments.
- Fiber optic splicing, laying, and certification.
- Electrical installations related to networks and telecommunications.

ETRA R&D (Internship) – Valencia

Website: www.grupoetra.com | **Business or sector:** Professional, scientific and technical activities

Data Engineer

[1 Jun 2022 – 31 Aug 2022]

During my time at ETRA R&D, I developed an internal tool designed to optimize the operational management of public bus fleets, with objectives such as efficiency, sustainability, and cost reduction.

My main contributions included:

- Processing operational data, such as energy consumption, geolocation, downtime, and charge status.
- Automation of data processing using Python scripts (decoding and cleaning) and structuring in SQL for subsequent analysis.
- Design and execution of workflows in Alteryx to filter, transform, and enrich information according to criteria such as line, time range, consumption type, and technical incidents.
- Strategic visualization of KPIs through interactive dashboards in Power BI, aimed at facilitating decision-making by operational and maintenance teams.
- Development of an internal application in C# to visualize operational bus data, using Jira and Confluence for task tracking and documentation.

This system significantly reduced the time spent on manual analysis, minimized operational errors, and provided key information for route optimization and maintenance planning.

EDUCATION AND TRAINING

Telecommunications Technology and Services Engineering

Polytechnical University of Valencia [Dec 2025]

City: Valencia | **Country:** Spain

- High Academic Performance Group (ARA)
- Specialization in Telematic Systems

- **Bachelor Thesis: Fraud Detection in FinTech Transactions using Artificial Intelligence**
  - Developed predictive models (Random Forest, XGBoost, LightGBM) with a focus on reducing financial losses in transaction-based environments.
  - Worked with highly imbalanced datasets, applying techniques such as SMOTE, class weighting and probability calibration to ensure reliable model behavior.
  - Optimized model decision thresholds based on economic impact and business logic, aligning technical outputs with real-world financial and regulatory constraints.

Erasmus+ Exchange Programme

Technical University of Munich [Oct 2024 – Aug 2025]

City: Munich | **Country:** Germany

LANGUAGE SKILLS

Spanish - Native | English - C1 | French - B2 | German - B1 | Valencian - C1

SKILLS

Business & Analytics

Power BI / Alteryx / Excel / SQL

Software Development

Python (pandas, sci-kit learn) / C# / Java

Engineering & Systems

MATLAB / C++ / Verilog / Cadence Virtuoso / Packet Tracer / Wireshark

Project & Collaboration Tools

Jira / Confluence / GitHub

VOLUNTEERING

[1 Jul 2024 – 1 Aug 2024] Accra, Ghana

Community Development Volunteer Soccer coach for kids from 6 to 14 years in Accra, Ghana.

Link: www.dreamafricacarefoundation.org

CONFERENCES AND SEMINARS

[1 Apr 2025 – 30 Jul 2025] Munich

Seminar Report: Hybrid Neural Networks This seminar paper reviews state-of-the-art Hybrid Neural Networks, combining artificial and spiking models to create more efficient and brain-inspired computing systems. It analyzes key approaches, evaluating their architecture, scalability, energy efficiency and future research directions.